



Institute of technology of Cambodia
Department GIC



Course Enrollment Project
I4-GIC-Group C
Individual Report

Name

KEO SIENGHENG

ID:

e20221161

Score:

.....

Software Engineering

Lecturer: Mr. ROEUN Pacharoth (TP)

Table Content

I. Introduction	1
I. My Role in the Project	1
II. Application Structure and Packages	1
III. Main Contributions	1
Application Startup	1
IV. Database Configuration	1
V. Security Implementation	2
VI. Password Handling and Authentication Utilities	2
VII. Auditing and Logging	2
VIII. Important Files	2
IX. Challenges and What I Learned	2

I. Introduction

This report explains my individual contribution to the Course Enrollment System project. The system was developed using Java Spring Boot with Gradle. My main responsibility was to build the foundation of the application. This included configuration, security, and core features that other parts of the system depend on. This foundation is important because it keeps the application stable, secure, and easy to maintain.

I. My Role in the Project

In this project, I was responsible for setting up the core part of the system. My work focused on how the application starts, how it connects to the database, and how security is handled. I also created shared utilities and logging features that support the rest of the system. These tasks helped ensure that other team members could develop their features without technical problems.

II. Application Structure and Packages

Most of my work was done in the core packages of the project. The main application file is `DemoGradleApplication.java`, which is used to start the Spring Boot application. I worked with the `config` package to handle system configuration and database settings. The `security` package was used to implement authentication and authorization. The `aspect` package was used for logging and auditing important system actions. The `util` package contains helper functions that are used in different parts of the application.

III. Main Contributions

Application Startup

I made sure that the Spring Boot application starts correctly using Gradle. I configured the project so that all components are detected and loaded properly. This allows the application to run smoothly in different environments.

IV. Database Configuration

At the beginning of the project, I tried to use Supabase as the database. However, I faced many problems such as connection errors and database constraint issues. Because of this, I decided to switch to MySQL or MariaDB. This database was easier to set up, more stable, and allowed

faster development. I configured the database connection and ensured that the application could store and retrieve data correctly.

V. Security Implementation

Security was one of the most important parts of my work. I implemented Spring Security to protect the application and its API endpoints. I used JWT authentication to secure user access and keep the system stateless. I also integrated OAuth2 so users could authenticate using external providers. In addition, I created custom authentication handlers to manage login success and failure in a controlled way.

VI. Password Handling and Authentication Utilities

To protect user data, I implemented password encoding so that passwords are stored securely in the database. I also created utility functions to help with authentication and token handling. This ensures that sensitive information is never stored or transmitted in plain text.

VII. Auditing and Logging

I developed an auditing feature using Aspect Oriented Programming. This allows the system to automatically log important actions such as authentication events and sensitive operations. This makes the system easier to monitor, debug, and maintain.

VIII. Important Files

Some of the key files I worked on include SecurityConfig.java, which controls the security configuration of the application. JwtTokenProvider.java is responsible for creating and validating JWT tokens. CustomUserDetailsService.java loads user information for authentication. AuditAspect.java handles logging and auditing across the system.

IX. Challenges and What I Learned

One of the biggest challenges was working with Supabase at the beginning of the project. Even though it caused delays, it helped me learn more about database integration and error handling. Switching to MySQL or MariaDB was an important decision that improved the stability of the project. Through this work, I gained a better understanding of Spring Boot, Spring Security, JWT authentication, and application design.

Conclusion

In conclusion, my contribution focused on building a strong and secure foundation for the Course Enrollment System. The configuration, security layer, and core features I implemented support all other modules in the application. This project improved my technical skills and gave me real experience in backend development using Java Spring Boot.